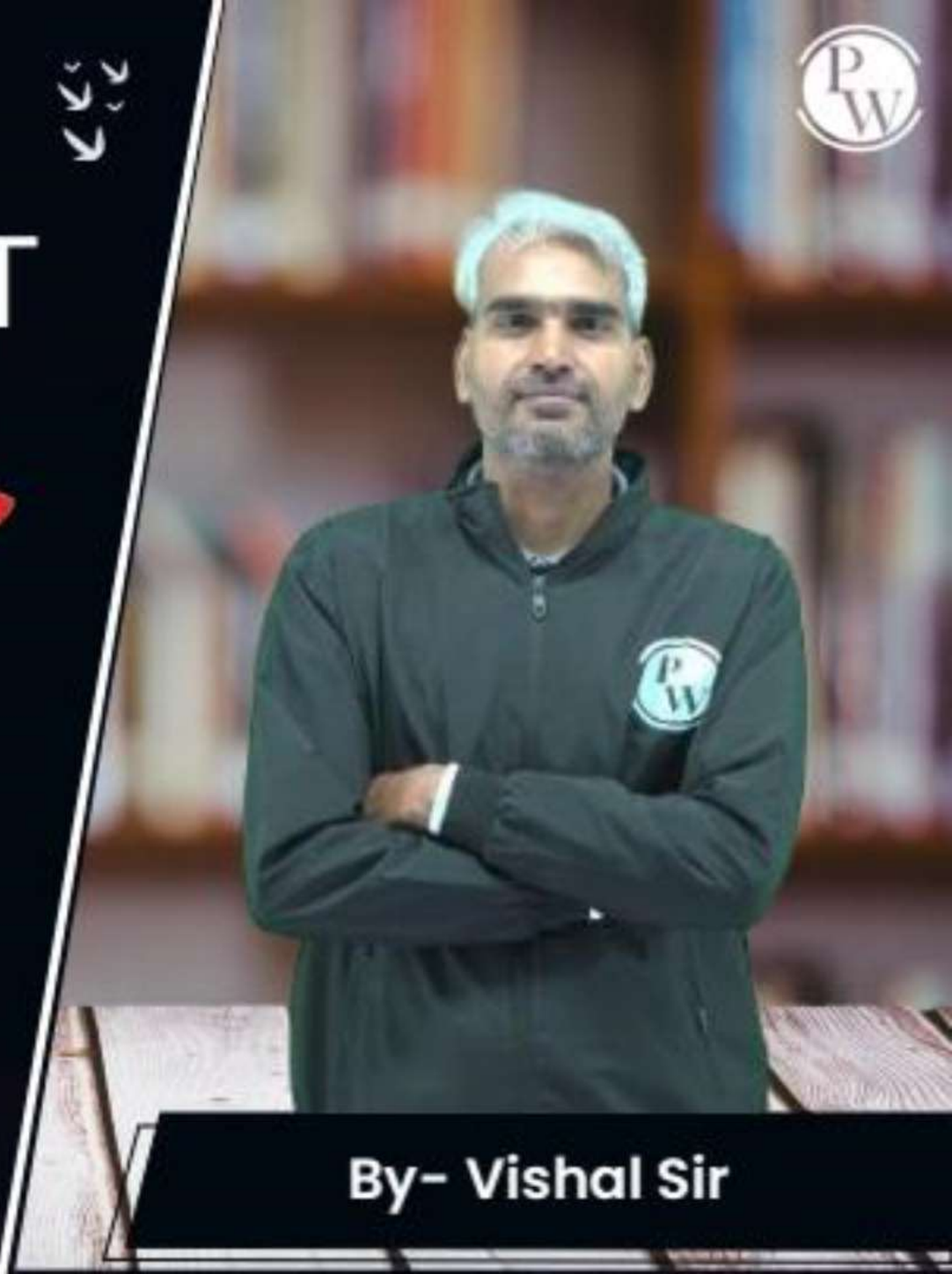
Computer Science & IT



Database Management System

Conversion from ER Model to Relational Model

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture

- Topic: ER model & ER diagram
- Topic: Integrity constraints



Slide

Topics to be Covered



- Topic: Integrity constraints
- Topic: Conversion from ER model to relational table

Slide



Topic : Relational Model concept

- The relational model was proposed by E.F. Codd to model data in the form of relations or tables.
- The relational model is represented by a table with columns and rows. Each row is known as a tuple.



Topic : Relational Constraints

- These are the restrictions or set of rules imposed on the database content. It validates the quality of the database. It validates that various operations like data insertion, updation, etc., can be performed without affecting the integrity of the data.

- Constraints could be either on a column level or a table level. The column level constraints are applied only to one column, whereas the table level constraints are applied to the whole table.



Topic : Types of Constraints on Relational Model

1. Domain constraint
2. Key constraint or Tuple Uniqueness constraint
3. Entity Integrity constraint
4. Referential Integrity constraint



Topic : Domain constraint

(attribute)

- Every domain must contain atomic values(smallest indivisible units) which means composite and multi-valued attributes are not allowed.

- We perform a datatype check here, which means when we assign a data type to a column we limit the values that it can contain. Eg. If we assign the datatype of attribute age as int, we can't give it values other than int datatype



Topic : Key constraint or Tuple Uniqueness constraint

- These are called uniqueness constraints since it ensures that every tuple in the relation should be unique.

tuple

{ i.e., we must be able to define candidate key for the database }

A relation can have multiple keys or candidate keys(minimal superkey), out of which we choose one of the keys as the primary key, we don't have any restriction on choosing the primary key out of candidate keys.



Topic : Entity Integrity constraint

- Entity integrity constraint specifies that no attribute of primary key must contain a null value in any relation.



Topic : Referential Integrity constraint

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The Referential integrity constraint is specified between two relations or tables and used to maintain the consistency among the tuples in two relations.

Some times may be on the same table



Topic : Referential Integrity constraint

- Foreign Key: A foreign key is a set of attributes in a table that refers to the primary key or alternate key of same table or some other table.

eg:

it references to the "Stu-id" attribute of Student relation

Foreign key

Primary key of student

Names of the attributes need not be same

Sid	Cid
S_1	C_1
S_1	C_2
S_3	C_2
S_4	C_3

Enroll

it is also a foreign key which will reference P.k. "Cid" of the course relation

(Relationship)

Enroll is the referencing relation

Stu-id	Sname	Branch
S_1		
S_2		
S_3		
S_4		

Student

{ Student is a referenced relation }

Cid	Cname	Branch
C_1		
C_2		
C_3		
C_4		

Course

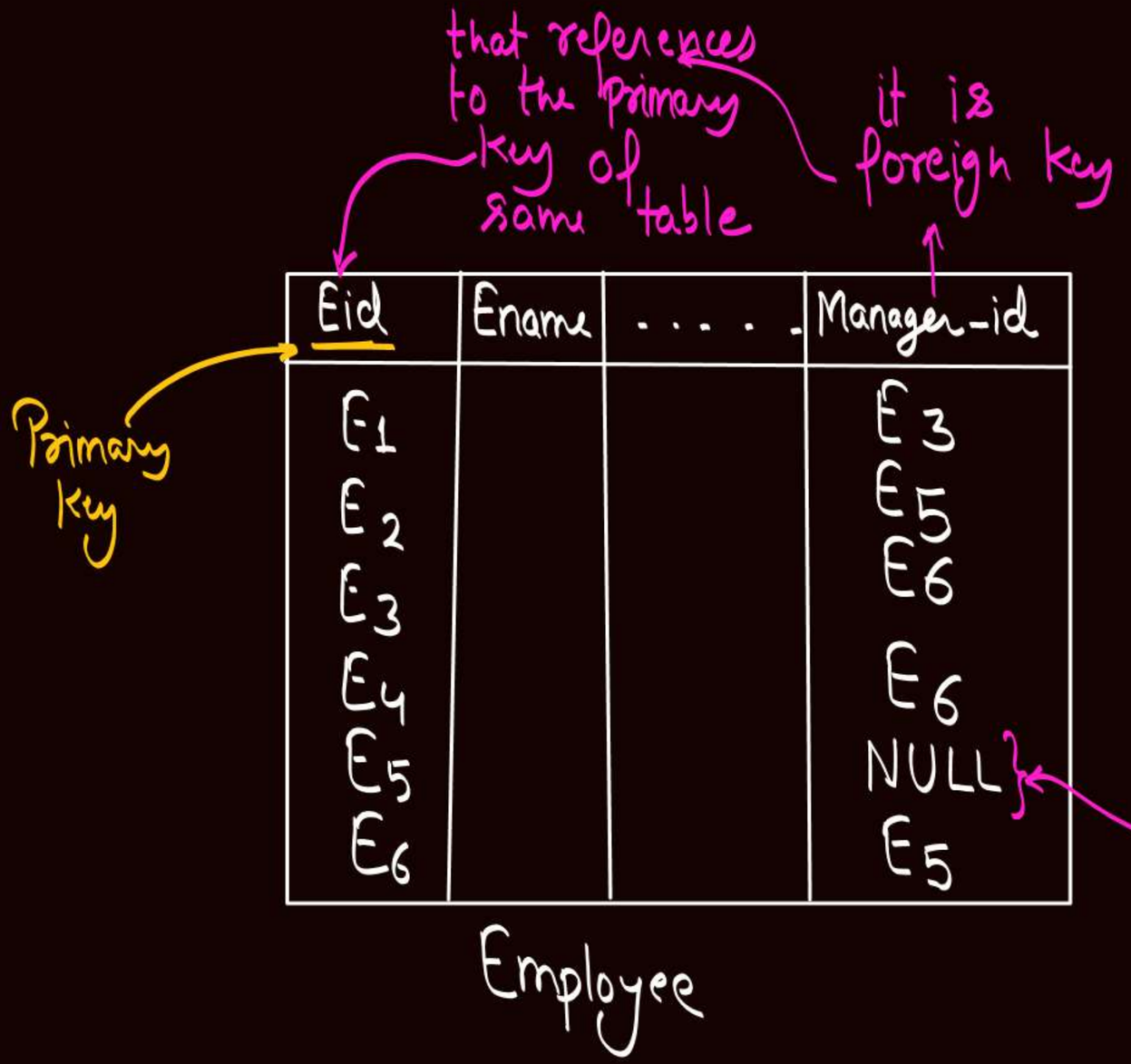
Note:

① A relation that contains foreign key is called a referencing relation

② Relation that contains the primary key / alternate key which is being referenced by the foreign key is called a referenced relation

In some cases, referencing relation & referenced relation may be same

eg:-



Employee relation is referencing relation as well as referenced relation

Foreign key attributes may take NULL value



Topic : Referential Integrity constraint

- Referential integrity constraint is enforced through a foreign key

∴ R_1 is referencing ∴ R_2 is Referenced.

- Let foreign key in relation R1 refers to primary key of relation R2.

The values of the foreign key in a tuple of relation R1 can either take the values from the primary key (values of) some tuple in relation R2, or can take NULL values, but can't be empty.

Values in foreign key Column $\subseteq$ [Values in Primary key/Alternate key column of referenced relation $\cup$ {NULL}]



Topic : Referential Integrity constraints on Referenced relation

- **Insertion:** Insertion of a tuple in referenced relation does not result in any referential integrity violation.

- **Deletion:** Deletion of a tuple from referenced relation may result in referential integrity violation.

- **Updation:** Updation of a tuple in referenced relation may result in referential integrity violation.



Topic : Referential Integrity constraints on Referencing relation

- **Insertion:** Insertion of a tuple in referencing relation may result in referential integrity violation.

- **Deletion:** Deletion of a tuple from referencing relation does not result in any referential integrity violation.

- **Updation:** Updation of a tuple in referencing relation may result in referential integrity violation.



Topic : Referential Integrity constraints on Referenced relation

(or Alternate key value) Because of deletion

- Deletion of a primary key value from the referenced relation may causes integrity violation. We may choose one of the following approaches to avoid integrity violation.

- On Delete No Action:
- On Delete Cascade:

On Delete Set NULL:

On delete No action

If deletion of a tuple from referenced relation causes any integrity violation, then that deletion is prohibited..

{ i.e. No action will be performed }

Not even the deletion from referenced relation

On delete Cascade

* If deletion of a tuple from referenced relation causes any integrity violation, then delete the tuple from the referenced relation, and also delete the corresponding tuples from referencing relation

it may result in deletion of few more tuples

it can happen when referenced relation and referencing relation are same

On delete Set NULL

* Case (1) If deletion of a tuple from referenced relation causes any integrity violation and foreign key attribute is allowed to take the NULL value, then delete the tuple from the referenced relation and set the value of the foreign key attribute as "NULL" in the tuples that causes integrity violation

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* Case (2) If deletion of a tuple from referenced relation causes any integrity violation and foreign key attribute is not allowed to take the NULL value, then deletion from the referenced relation itself is prohibited, i.e; if foreign key attribute is not allowed to take NULL values then "On delete Set NULL" is same as "On delete no action"



Topic : Referential Integrity constraints on Referenced relation

(or Alternate key value) "Because of updation"

- Updation in the primary key value in the referenced relation may causes integrity violation. We may choose one of the following approaches to avoid integrity violation.

- On Update No Action:
- On Update Cascade:
- On Update Set NULL:

On Update No action

If updation of a tuple in referenced relation causes any integrity violation, then that updation is prohibited.

{ i.e. No action will be performed }

Not even the updation in referenced relation

On Update Cascade

If updation of a tuple in referenced relation causes any integrity violation, then update the tuple in the referenced relation, and also update the values in the foreign key column of the referencing relation

On Update Set NULL

- Case (1) If updation of a tuple in referenced relation causes any integrity violation and foreign key attribute is allowed to take the NULL value, then update the tuple in the referenced relation and set the value of the foreign key attribute as "NULL" in the tuples that causes integrity violation

---

Case (2) If updation of a tuple in referenced relation causes any integrity violation and foreign key attribute is not allowed to take the NULL value, then updation in the referenced relation itself is prohibited, i.e; if foreign key attribute is not allowed to take NULL values then "On Update Set NULL" is same as "On Update no action"



Topic : Referential Integrity constraints on Referencing relation

- **Insertion:** Insertion of a tuple in referencing relation may result in referential integrity violation.

Deletion: Deletion of a tuple from referencing relation does not result in any referential integrity violation.

Updation: Updation of a tuple in referencing relation may result in referential integrity violation.

Referenced relation is Parent relation

whereas

Referencing relation is Child relation



Topic : Referential Integrity constraints on Referencing relation

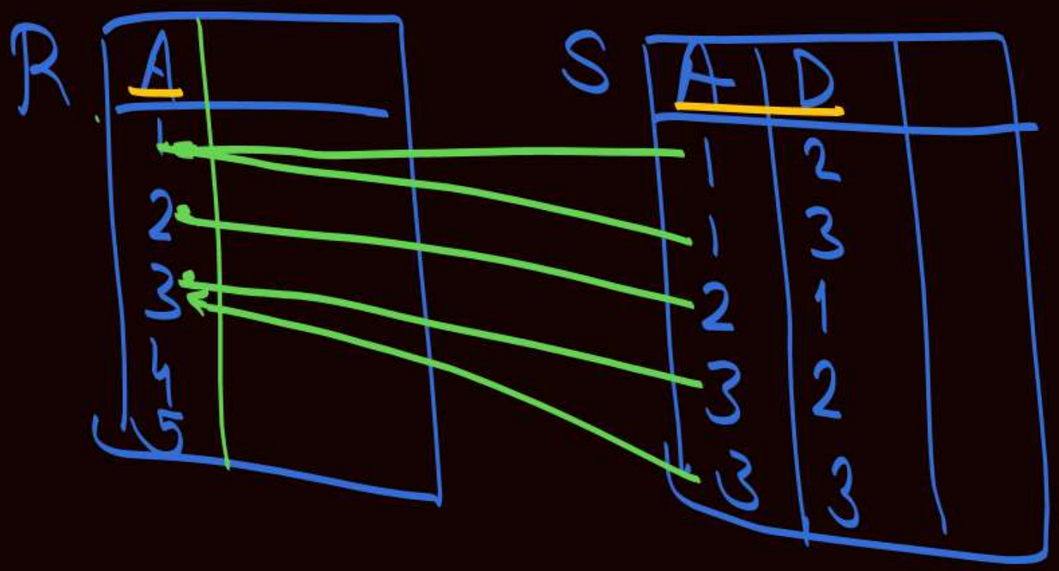
i.e; on child relation

If any operation in referencing relation (child relation) causes any referential integrity violation, then corresponding operation is prohibited.

eg:- Let R(A B C) is a relation with 50 tuples,
and S(A D E) is a relation with 25 tuples,

then number of tuples in R ⋈ S = ?

{ Attribute 'A' in relation 'S' references to the Primary key 'A' of relation R }



Number of tuples in R⋈S : 25

Same as the no. of tuples in relation with foreign key

eg:- Let R(A B C) is a relation with 50 tuples, and S(A D E) is a relation with 25 tuples, then number of tuples in R ⋈ S = ?

{ Attribute 'A' in relation 'S' references to the Primary key 'A' of relation R }

R

A
1
2
3
4
5

S

A	D	E
NULL	2	1
1	3	2
NULL	1	3
2	2	4
3	3	3

Number of tuples in R ⋈ S

- Minimum # tuples = 0
- Maximum # tuples = 25

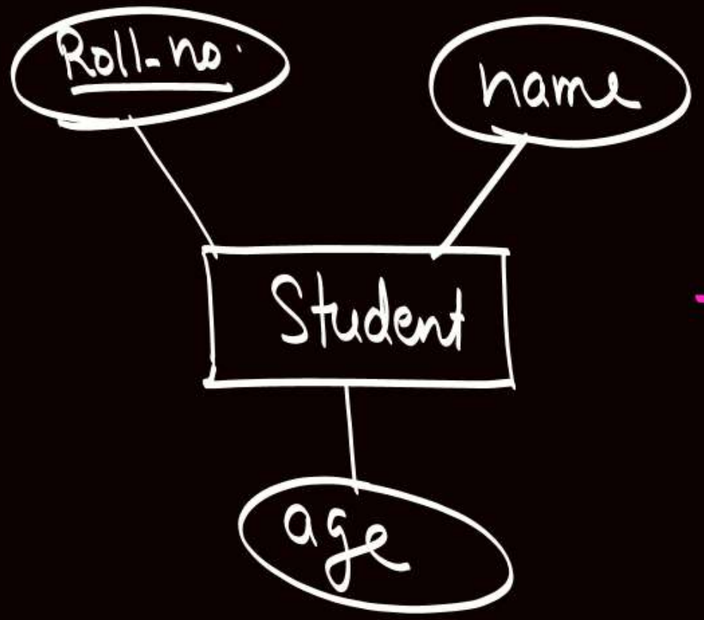
ER Model to Relational Table

Objective :-

Given an ER diagram, we need to Convert it into relational model, using minimum number of relational table.

- Given an entity set, we will define relation w.r.t. that entity set.

① All attributes are Simple attributes: -



ER model

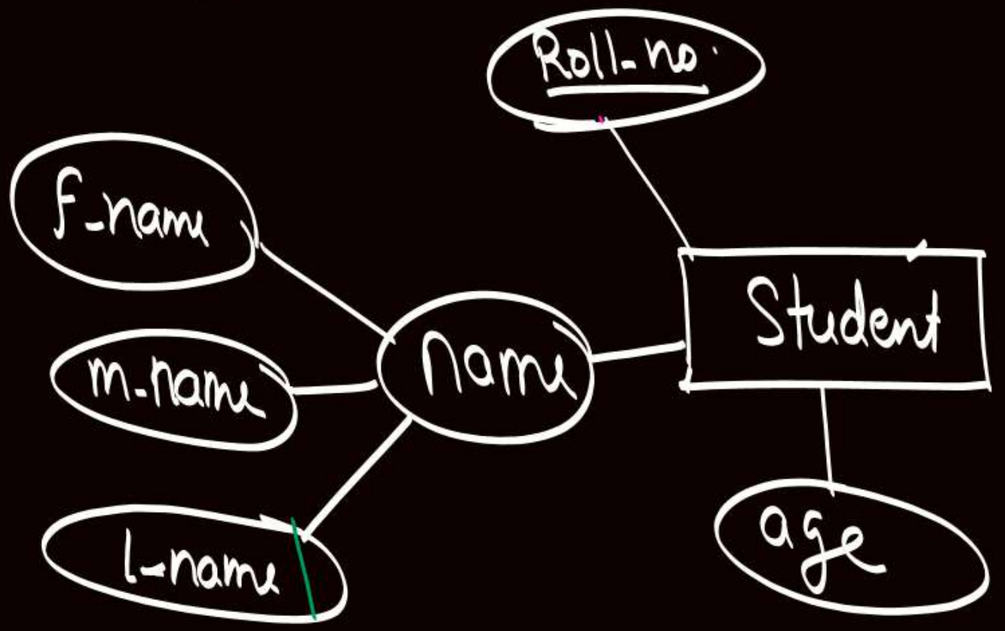
Relational table

Student

Roll_no	name	age

* Given an entity set, we will define relation w.r.t. that entity set.

② When some attributes are Composite attributes :-



ER model

Relational table

In the relational table we will consider only simple components of Composite attribute

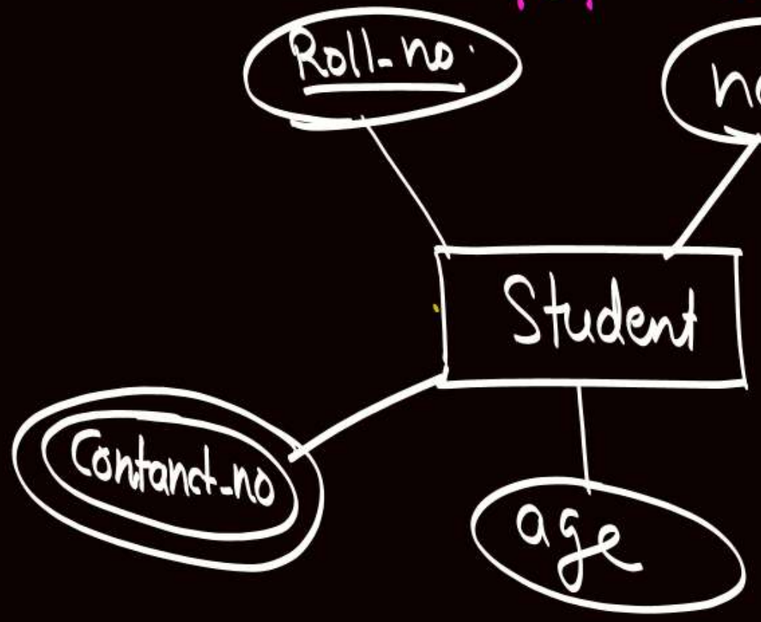
Student

Roll_no	f_name	m_name	l_name	age

* Given an entity set, we will define relation w.r.t. that entity set.

③ Some attributes are multivalued attributes :

101 94779, 94250



ER model

⇒ Relational tables ⇒ Two relational tables

One, w.r.t. all simple attributes

Student

Roll-no.	name	age

one w.r.t multivalued attribute "Contact-no"

Stu-Contact-info

Roll-No.	Contact-no
101	94278
101	99930
102	98270

{ for each mult-valued attribute, we will require a seperate relation }

Note: There is no representation of Composite attributes and multivalued attributes in relational model



2 mins Summary

Topic Integrity constraints

Topic Conversion from ER model to relational table

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THANK - YOU